

Internship Project Phase 1 Report (Blue Print)

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Project: E-Commerce Sales Performance Dashboard with Cohort Analysis

External guide:

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1. Introduction

In the modern digital economy, e-commerce businesses generate large volumes of transactional data. However, without structured analysis, this data does not translate into actionable insights. Organizations often struggle to understand customer behavior, retention patterns, and sales performance trends.

This project aims to develop an **E-Commerce Sales Performance Dashboard integrated with Cohort Analysis** to transform raw transactional data into meaningful business insights. The solution will enable stakeholders to make informed decisions regarding customer retention, product performance, and revenue optimization.

2. Problem Statement

E-commerce businesses lack a comprehensive analytical system to monitor customer retention and sales performance effectively. This results in limited visibility into repeat customer behaviour, revenue drivers, and long-term customer value, thereby affecting strategic decision-making.

3. Objectives

The objectives of this project are as follows:

- To analyze **overall sales performance** and revenue trends
- To identify **top-performing products and regions**
- To study **customer behavior and purchasing patterns**
- To perform **cohort analysis** to evaluate customer retention over time
- To design and develop an **interactive dashboard using Power BI**
- To generate **data-driven insights and recommendations**

4. Dataset Description

The project utilizes the **Online Retail II dataset**, which contains transactional data from a UK-based online retail company.

- **Time Period:** December 2009 – December 2011
- **Business Type:** Non-store online retail (gift items)
- **Customer Type:** Primarily wholesalers
- **Data Nature:** Transaction-level data

Key Attributes:

- Invoice Number (transaction identifier; includes cancellations)
- Product Code and Description
- Quantity of items purchased
- Invoice Date and Time
- Unit Price (in GBP)
- Customer ID
- Country

The dataset initially contains **5,41,910 transactions**, which are later cleaned and reduced for accurate analysis.

5. Scope of the Project

The scope of this project includes the following analytical components:

5.1 Sales Performance Analysis

- Revenue trends over time (monthly and yearly)
- Identification of peak sales periods
- Overall business growth patterns

5.2 Product Analysis

- Identification of top-selling products
- Revenue contribution by product categories
- Product demand distribution

5.3 Customer Analysis

- Total customer count
- New vs returning customers
- Purchase frequency and behavior

5.4 Geographic Analysis

- Country-wise sales distribution
- Identification of key revenue-generating regions

5.5 Cohort Analysis

- Grouping customers based on first purchase month
- Tracking retention over subsequent periods
- Understanding customer lifecycle and churn patterns

6. What Can Be Done Using This Dataset

The dataset enables the following analytical possibilities:

- Time-series analysis of revenue and transactions
- Identification of customer retention and churn trends
- Regional performance comparison
- Creation of cohort-based retention matrices

7. What Will Be Done in This Project

The project will specifically implement the following:

- Data cleaning and preprocessing using Python
- Removal of cancellations, missing values, and invalid records
- Creation of derived features such as revenue and time-based attributes
- Exploratory Data Analysis (EDA) with visualizations
- Cohort analysis to calculate customer retention rates
- Development of a Power BI dashboard with key performance indicators

8. Hypothesis

The project is based on the following assumptions:

1. Repeat customers are expected to contribute approximately 60–70% of total revenue
2. Customer retention decreases after the first purchase month
3. A small percentage of products generate the majority of revenue
4. Certain geographic regions contribute more to overall sales

9. Methodology

The project follows a structured workflow to transform raw data into actionable insights

Data Collection → Data Cleaning → Feature Engineering → Exploratory Data Analysis → Cohort Analysis → Dashboard Development

- Initially, the raw dataset is collected and cleaned to remove invalid and inconsistent records.
- Next, relevant features such as revenue and time-based attributes are created.
- Exploratory Data Analysis is then performed to identify patterns and trends.
- Cohort analysis is applied to study customer retention behavior over time.
- Finally, the insights are visualized through an interactive Power BI dashboard for business decision-making.

10. Tools and Technologies

- **Python** (Pandas, Matplotlib, Seaborn) - used for data cleaning, analysis, and visualization
- **Jupyter Notebook** - used for step-by-step implementation and experimentation
- **Power BI** - used for building an interactive and business-friendly dashboard

11. Resources Used

The following resources will support the project:

- Online Retail II Dataset (UCI / Kaggle)
- Python documentation and data analysis libraries
- Power BI Desktop
- Online tutorials and reference materials
- ChatGPT

12. Project Plan (2-Day Execution)

Day 1: Data Preparation and Analysis

- Perform data cleaning and preprocessing
- Handle missing values and cancellations
- Create revenue and time-based features

- Conduct exploratory data analysis
- Generate initial insights and visualizations

Day 2: Cohort Analysis and Dashboard Development

- Perform cohort analysis and retention calculation
- Create retention matrix
- Export processed data
- Build Power BI dashboard
- Finalize insights and prepare presentation

13. Expected Outcomes

The project is expected to deliver:

- A fully functional **interactive dashboard**
- A **cohort retention matrix** showing customer behavior
- Clear insights into **sales trends and product performance**
- Identification of **key revenue drivers**
- Customer segmentation using RFM analysis
- Actionable recommendations for business improvement

14. Conclusion

This blueprint outlines a structured approach to developing a comprehensive e-commerce analytics solution. By combining sales performance analysis with cohort-based customer retention analysis, the project aims to deliver valuable insights that support data-driven decision-making and business growth.